1 Fundamentals

$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right]=\frac{g(x)f'(x)-f(x)g'(x)}{(g(x))^2}$

- **Normal**: $\frac{\exp(-\frac{1}{2}(\mathbf{x}-\mu)^T\Sigma^{-1}(\mathbf{x}-\mu))}{\sqrt{(2\pi)^k\det(\Sigma)}}$
- **Beta**: $\mathrm{Beta}(\theta;\alpha,\beta)\propto\theta^{\alpha-1}(1-\theta)^{\beta-1}$
- **Laplace**: $\frac{1}{2l}\exp(-\frac{|x-\mu|}{l})$

- Gaussian CDF has no closed-form; $O(n^2)$ params.

- $\mathbb{E}[\mathbf{AX}+\mathbf{b}]=\mathbf{A}\mathbb{E}[\mathbf{X}]+\mathbf{b}$; $\mathbb{E}[\mathbf{X}+\mathbf{Y}]=\mathbb{E}[\mathbf{X}]+\mathbb{E}[\mathbf{Y}]$
- $\mathbb{E}[\mathbf{XY}^\top]=\mathbb{E}[\mathbf{X}]\cdot\mathbb{E}[\mathbf{Y}]^\top$ (if $\mathbf{X}$, $\mathbf{Y}$ indep.)
- LOTUS: $\mathbb{E}[\mathbf{g}(\mathbf{X})]=\int_{\mathbf{X}(\Omega)}\mathbf{g}(\mathbf{x})\cdot p(\mathbf{x})d\mathbf{x}$ (if $\mathbf{g}$ nice and $\mathbf{X}$ cont.)
- Tower rule: $\mathbb{E}_{\mathbf{Y}}[\mathbb{E}_{\mathbf{X}}[\mathbf{X}|\mathbf{Y}]]=\mathbb{E}[\mathbf{X}]$

- $\mathrm{Var}[\mathbf{X}]\doteq\mathbb{E}[(\mathbf{X}-\mathbb{E}[\mathbf{X}])(\mathbf{X}-\mathbb{E}[\mathbf{X}])^\top]$ $=\mathbb{E}[\mathbf{XX}^\top]-\mathbb{E}[\mathbf{X}]\cdot\mathbb{E}[\mathbf{X}]^\top=\mathrm{Cov}[\mathbf{X},\mathbf{X}]$
- $\mathrm{Var}[\mathbf{AX}+\mathbf{b}]=\mathbf{A}\mathrm{Var}[\mathbf{X}]\mathbf{A}^\top$
- $\mathrm{Var}[\mathbf{X}+\mathbf{Y}]=\mathrm{Var}[\mathbf{X}]+\mathrm{Var}[\mathbf{Y}]+2\mathrm{Cov}[\mathbf{X},\mathbf{Y}]$
- $\mathrm{Var}[\mathbf{X}+\mathbf{Y}]=\mathrm{Var}[\mathbf{X}]+\mathrm{Var}[\mathbf{Y}]$ (if $\mathbf{X}$, $\mathbf{Y}$ indep.)
- Law of total variance, LOTV: $\mathrm{Var}[\mathbf{X}]=\mathbb{E}_{\mathbf{Y}}[\mathrm{Var}_{\mathbf{X}}[\mathbf{X}|\mathbf{Y}]]+\mathrm{Var}_{\mathbf{Y}}[\mathbb{E}_{\mathbf{X}}[\mathbf{X}|\mathbf{Y}]]$

- $\mathrm{Cov}[\mathbf{X},\mathbf{Y}]\doteq\mathbb{E}[(\mathbf{X}-\mathbb{E}[\mathbf{X}])(\mathbf{Y}-\mathbb{E}[\mathbf{Y}])^\top]$ $=\mathbb{E}[\mathbf{XY}^\top]-\mathbb{E}[\mathbf{X}]\cdot\mathbb{E}[\mathbf{Y}]^\top$
- $\mathrm{Cov}[\mathbf{X},\mathbf{Y}]=\mathrm{Cov}[\mathbf{Y},\mathbf{X}]$; $\mathrm{Cov}[\mathbf{X},\mathbf{X}]\geq\mathbf{0}$
- $\mathrm{Cov}[\mathbf{AX}+\mathbf{c},\mathbf{BY}+\mathbf{d}]=\mathbf{A}\mathrm{Cov}[\mathbf{X},\mathbf{Y}]\mathbf{B}^\top$

- *Correlation* is normalized covariance: $\mathrm{Cor}[\mathbf{X},\mathbf{Y}](i,j)\doteq\frac{\mathrm{Cov}[X_i,Y_j]}{\sqrt{\mathrm{Var}[X_i]\mathrm{Var}[Y_j]}}\in[-1,1]$
- *Uncorrelated* iff $\mathrm{Cov}[\mathbf{X},\mathbf{Y}]=\mathbf{0}$.
- **Change of variables:** Let $\mathbf{g}$ be diff. and inv. Then for $\mathbf{Y}=\mathbf{g}(\mathbf{X})$: $p_{\mathbf{Y}}(\mathbf{y})=p_{\mathbf{X}}(\mathbf{g}^{-1}(\mathbf{y}))\cdot|\det(\mathbf{Dg}^{-1}(\mathbf{y}))|$ where $\mathbf{Dg}^{-1}(\mathbf{y})$ is the Jacobian of $\mathbf{g}^{-1}$ at $\mathbf{y}$.

Bayes' rule: $p(\mathbf{x}|\mathbf{y})=\frac{p(\mathbf{y}|\mathbf{x})\cdot p(\mathbf{x})}{p(\mathbf{y})}$

- If prior $p(\mathbf{x})$ and posterior $p(\mathbf{x}|\mathbf{y})$ from same fam. of distr., prior is **conjugate prior** to likelihood $p(\mathbf{y}|\mathbf{x})$.
- Under some conditions, **Gaussian is self-conjugate** (Gaussian prior and likelihood $\rightarrow$ posterior Gaussian).
- Gaussians with known var. are self-conjugate to any Gaussian likelihood.

- Choosing non-informative prior in absence of evidence is **principle of indifference/insufficient reason**.
- **Improper prior:** not required that prior is a valid distr. (i.e., integrates to 1). Can still derive meaning.
- **Max. entropy principle:** choose prior s.t. one that makes the least "additional assumptions", i.e., prior least "informative".

Gaussian properties

- **Gaussians have max. entropy among all distr.** with known mean and var.: $1/2\cdot\log((2\pi e)^d\det(\mathbf{\Sigma}))$
- Jointly Gaussian random vectors, $\mathbf{X}$ and $\mathbf{Y}$, are independent iff $\mathbf{X}$ and $\mathbf{Y}$ are uncorrelated.
- Closed under marginalization and conditioning.

Let $\mathbf{X}$ be Gaussian and index sets $A,B\subseteq[n]$. For any **marginal distr.** $\mathbf{X}_A\sim\mathcal{N}(\mu_A,\mathbf{\Sigma}_{AA})$ and for any **conditional distr.**: $\mathbf{X}_A|\mathbf{X}_B=\mathbf{x}_B\sim\mathcal{N}(\mu_{A|B},\mathbf{\Sigma}_{A|B})$ where $\mu_{A|B}\doteq\mu_A+\mathbf{\Sigma}_{AB}\mathbf{\Sigma}_{BB}^{-1}(\mathbf{x}_B-\mu_B)$, $\mathbf{\Sigma}_{A|B}\doteq\mathbf{\Sigma}_{AA}-\mathbf{\Sigma}_{AB}\mathbf{\Sigma}_{BB}^{-1}\mathbf{\Sigma}_{BA}$. Observe that the variance can only shrink.

- Additive and closed under affine transformations.

- $M\cdot\mathcal{N}(\mu,\mathbf{\Sigma})=\mathcal{N}(M\mu,M^\top\mathbf{\Sigma}M)$
- $\mathcal{N}(\mu_A,\mathbf{\Sigma}_A)+\mathcal{N}(\mu_B,\mathbf{\Sigma}_B)=\mathcal{N}(\mu_A+\mu_B,\mathbf{\Sigma}_A+\mathbf{\Sigma}_B)$
- $\mathcal{N}(\mu_A,\mathbf{\Sigma}_A)\cdot\mathcal{N}(\mu_B,\mathbf{\Sigma}_B)\propto\mathcal{N}(\cdot,\cdot)$

Maximum likelihood estimate (MLE): $\hat\theta_{\mathrm{MLE}}\doteq\underset{\theta\in\Theta}{\mathrm{argmax}}\,p(y_{1:n}|\mathbf{x}_{1:n},\theta)=\underset{\theta\in\Theta}{\mathrm{argmax}}\sum_{i=1}^n\log p(y_i|\mathbf{x}_i,\theta)$

- **Consistent** if: $\hat\theta_{\mathrm{MLE}}\xrightarrow{\mathbb{P}}\theta^\star$ as $n\rightarrow\infty$.
- **Asymptotically normal** if $\hat\theta_{\mathrm{MLE}}\xrightarrow{\mathcal{D}}\mathcal{N}(\theta^\star,\mathbf{S}_n)$ as $n\rightarrow\infty$ where $\mathbf{S}_n$ is asymptotic covar. of MLE.
- MLE is **asymptotically efficient** (there exists no other consistent estimator with a "smaller" asymptotic var.).
- For the finite sample regime, the MLE need not be unbiased, and it is susceptible to overfitting to the (finite) training data.

Maximum a posterior (MAP) estimate: $\hat\theta_{\mathrm{MAP}}\doteq\mathrm{argmin}_{\theta\in\Theta}\underbrace{-\log p(\theta)}_{\text{regularization}}+\underbrace{\ell_{\mathrm{nll}}(\theta;\mathcal{D}_n)}_{\text{quality of fit}}$

The **log-prior** $\log p(\theta)$ acts as a regularizer. Common:

- $p(\theta)=\mathcal{N}(\theta;\mathbf{0},\lambda\mathbf{I})$ gives $-\log p(\theta)=\frac{\lambda}{2}\|\theta\|_2^2+\mathrm{const}$
- $p(\theta)=\mathrm{Laplace}(\theta;\mathbf{0},\lambda)$ gives $-\log p(\theta)=\lambda\|\theta\|_1+\mathrm{const}$
- Uniform prior gives const (no reg., MAP is MLE)

2 Bayesian Linear Regression (BLR)

$\hat{\mathbf{w}}_{\mathrm{ls}}=(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$; $\hat{\mathbf{w}}_{\mathrm{ridge}}=(\mathbf{X}^\top\mathbf{X}+\lambda\mathbf{I})^{-1}\mathbf{X}^\top\mathbf{y}$

- BLR can be viewed as GP regression with zero mean function and linear kernel function.

Gaussian prior on weights $\mathbf{w}\sim\mathcal{N}(\mathbf{0},\sigma_{\mathrm{p}}^2\mathbf{I})$:

- Yields Gaussian posterior $\mathbf{w}|\mathbf{x}_{1:n},y_{1:n}\sim\mathcal{N}(\mu,\mathbf{\Sigma})$, as $\log p(\mathbf{w}|\mathbf{x}_{1:n},y_{1:n})=-\frac{1}{2}[\mathbf{w}^\top\mathbf{\Sigma}^{-1}\mathbf{w}-2\mu]+\mathrm{const}$, w. $\mathbf{\Sigma}\doteq(\sigma_{\mathrm{n}}^{-2}\mathbf{X}^\top\mathbf{X}+\sigma_{\mathrm{p}}^{-2}\mathbf{I})^{-1}$ and $\mu\doteq\sigma_{\mathrm{n}}^{-2}\mathbf{\Sigma}\mathbf{X}^\top\mathbf{y}$.
- MAP is *identical to ridge regression* with $\lambda\doteq\sigma_{\mathrm{n}}^2/\sigma_{\mathrm{p}}^2$.
- **Bayesian inference**: Distr. for a test point $\mathbf{x}^\star$ is: $y^\star|\mathbf{x}^\star,\mathbf{x}_{1:n},y_{1:n}\sim\mathcal{N}(\mu^\top\mathbf{x}^\star,\mathbf{x}^{\star\top}\mathbf{\Sigma}\mathbf{x}^\star+\sigma_{\mathrm{n}}^2)$

Laplace prior on weights $\mathbf{w}\sim\mathrm{Laplace}(\mathbf{0},h)$:

- MAP is *identical to lasso regression* with $\lambda\doteq\sigma_{\mathrm{n}}^2/h$.

Heteroscedastic noise ϵ_i may depend on $\mathbf{x}_i$, while **Homoscedastic** may not.

$\mathrm{Var}[y^\star|\mathbf{x}^\star]=\underbrace{\mathbb{E}_\theta[\mathrm{Var}_{y^\star}[y^\star|\mathbf{x}^\star,\theta]]}_{\text{aleatoric uncertainty}}+\underbrace{\mathrm{Var}_\theta[\mathbb{E}_{y^\star}[y^\star|\mathbf{x}^\star,\theta]]}_{\text{epistemic uncertainty}}$

Aleatoric: noise in data; **Epistemic:** noise in model.

Applying linear reg. to non-linear fns.: use non-linear transformation ϕ to $\mathbf{X}$. Define $\mathbf{\Phi}=\phi(\mathbf{X})$. With Gaussian prior and $\mathbf{K}=\sigma_{\mathrm{p}}^2\mathbf{\Phi}\mathbf{\Phi}^\top$: $\mathbf{f}|\mathbf{X}\sim\mathcal{N}(\mathbf{\Phi}\mathbb{E}[\mathbf{w}],\mathbf{\Phi}\mathrm{Var}[\mathbf{w}]\mathbf{\Phi}^\top)=\mathcal{N}(\mathbf{0},\mathbf{K})$.

Kernel: $k(\mathbf{x},\mathbf{x}')\doteq\sigma_{\mathrm{p}}^2\phi(\mathbf{x})^\top\phi(\mathbf{x}')=\mathrm{Cov}[f(\mathbf{x}),f(\mathbf{x}')]$

- Choice of kernel implicitly determines the function class that $\mathbf{f}$ is sampled from, which encodes our prior beliefs.
- Kernel matrix has shape $n\times n$ (input space dimension) instead of $e\times e$ (feature space dimension).

For **inference**, define $\tilde{\mathbf{\Phi}}\doteq\begin{bmatrix}\mathbf{\Phi}\\ \phi(\mathbf{x}^{\star\top})\end{bmatrix}$, $\tilde{\mathbf{y}}\doteq\begin{bmatrix}\mathbf{y}\\ y^\star\end{bmatrix}$, $\tilde{\mathbf{f}}\doteq\begin{bmatrix}\mathbf{f}\\ f^\star\end{bmatrix}$. For $\tilde{\mathbf{f}}=\tilde{\mathbf{\Phi}}w$ we have: $\tilde{\mathbf{y}}|\mathbf{X},\mathbf{x}^\star\sim\mathcal{N}(\mathbf{0},\tilde{\mathbf{K}}+\sigma_n^2\mathbf{I})$

- **Linear**: $k(\mathbf{x},\mathbf{x}')=l\mathbf{x}^\top\mathbf{x}'$
- **RBF/Gaussian**: $k(\mathbf{x},\mathbf{x}')=\exp(-\frac{(\mathbf{x}-\mathbf{x}')^2}{2l^2})$ (larger l gives smoother fns.; cannot model under weight-space view of BLR; feature space are poly. of infinite degree)
- **Polynomial** $k(\mathbf{x},\mathbf{x}')=(1+\mathbf{x}^\top\mathbf{x}')^d$ (feature space are poly. of degree d)
- **Laplacian**: $k(\mathbf{x},\mathbf{x}')=\exp(-\alpha\|\mathbf{x}-\mathbf{x}'\|)$
- **Matérn**: (For $v=\frac{1}{2}$ Laplace, for $v\rightarrow\infty$ RBF)

Properties of kernels

- Symmetric: $k(\mathbf{x},\mathbf{x}')=k(\mathbf{x}',\mathbf{x})$
- $\mathbf{K}_{AA}$ is p.s.d.
- **Stationary** if there exists $\tilde k$ s.t. $\tilde k(\mathbf{x}-\mathbf{x}')=k(\mathbf{x},\mathbf{x}')$ (only relative location of points matters)
- **Isotropic** if there exists $\tilde k$ s.t. $\tilde k(\|\mathbf{x}-\mathbf{x}'\|_2)=k(\mathbf{x},\mathbf{x}')$ (only distance between points matters)

Composition of kernels

- Addition: $k(\mathbf{x},\mathbf{x}')\doteq k_1(\mathbf{x},\mathbf{x}')+k_2(\mathbf{x},\mathbf{x}')$ (OR)
- Multiplication: $k(\mathbf{x},\mathbf{x}')\doteq k_1(\mathbf{x},\mathbf{x}')\cdot k_2(\mathbf{x},\mathbf{x}')$ (AND)
- Mult. with const.: $k(\mathbf{x},\mathbf{x}')\doteq c\cdot k_1(\mathbf{x},\mathbf{x}')$ for any $c\geq0$
- Composition. with poly.: $k(\mathbf{x},\mathbf{x}')\doteq f(k_1(\mathbf{x},\mathbf{x}'))$ for any poly. f with positive coefficients.
- Composition. with exp.: $k(\mathbf{x},\mathbf{x}')\doteq\exp(k_1(\mathbf{x},\mathbf{x}'))$

Efficient online BLR ($O(d^2)$ instead of $O(d^3)$):

- $\mathbf{X}^{(t+1)\top}\mathbf{X}^{(t+1)}=\mathbf{X}^{(t)\top}\mathbf{X}^{(t)}+\mathbf{x}^{(t)\top}\mathbf{x}^{(t)}\in\mathbb{R}^{d\times d}$
- $\mathbf{X}^{(t+1)\top}\mathbf{y}^{(t+1)}=\mathbf{X}^{(t)\top}\mathbf{y}^{(t)}+\mathbf{x}^{(t)\top}\mathbf{y}^{(t)}\in\mathbb{R}^d$
- Since $\mathbf{X}^\top\mathbf{X}=\sum_{i=1}^t\mathbf{x}_i\mathbf{x}_i^\top$ and $\mathbf{X}^\top\mathbf{y}=\sum_{i=1}^t y_ix_i$

Logistic BLR:

$\hat{\mathbf{w}}_{\mathrm{MAP}}=\underset{\mathbf{w}\in\mathbb{R}^d}{\mathrm{argmin}}\frac{1}{2\sigma_{\mathrm{p}}^2}\|\mathbf{w}\|_2^2+\sum_{i=1}^n\log(1+\exp(-y_i\mathbf{w}^\top\mathbf{x}_i))$

- For $\lambda=1/(2\sigma_{\mathrm{p}}^2)$ this is equiv. to standard logistic reg. where $\ell_{\log}(\mathbf{w}^\top\mathbf{x};y)\doteq\log(1+\exp(-y\mathbf{w}^\top\mathbf{x}))$ and $\nabla_{\mathbf{w}}\ell_{\log}(\mathbf{w}^\top\mathbf{x};y)=-y\mathbf{x}\cdot\sigma(-y\mathbf{w}^\top\mathbf{x})$.
- Post. not Gaussian or closed, but log. density is convex

3 Gaussian Processes (GPs)

Mean fn. $\mu:\mathcal{X}\rightarrow\mathbb{R}$; **Covar. fn.** $k:\mathcal{X}\times\mathcal{X}\rightarrow\mathbb{R}$;

Using homoscedastic noise assumption: $y^\star|\mathbf{x}^\star,\mu,k\sim\mathcal{N}(\mu(\mathbf{x}^\star),k(\mathbf{x}^\star,\mathbf{x}^\star)+\sigma_{\mathrm{n}}^2)$

New point: Joint distribution of the observations $y_{1:n}$ and the noise-free prediction $f^\star$ at a test point $\mathbf{x}^\star$ as $\begin{bmatrix}\mathbf{y}\\ f^\star\end{bmatrix}|\mathbf{x}^\star,\mathbf{x}_{1:n}\sim\mathcal{N}(\tilde\mu,\tilde{\mathbf{K}})$ where $\tilde\mu\doteq\begin{bmatrix}\mu_A\\ \mu(\mathbf{x}^\star)\end{bmatrix}$, $\tilde{\mathbf{K}}\doteq\begin{bmatrix}\mathbf{K}_{AA}+\sigma_n^2\mathbf{I} & \mathbf{k}_{\mathbf{x}^\star,A}\\ \mathbf{k}_{\mathbf{x}^\star,A}^\top & k(\mathbf{x}^\star,\mathbf{x}^\star)\end{bmatrix}$, $\mathbf{k}_{\mathbf{x},A}\doteq[k(\mathbf{x},\mathbf{x}_1)\ldots k(\mathbf{x},\mathbf{x}_n)]^\top$

GP posterior update: $f|\mathbf{x}_{1:n},y_{1:n}\sim\mathcal{GP}(\mu',k')$ where $\mu'(\mathbf{x})\doteq\mu(\mathbf{x})+\mathbf{k}_{\mathbf{x},A}^\top(\mathbf{K}_{AA}+\sigma_{\mathrm{n}}^2\mathbf{I})^{-1}(\mathbf{y}_A-\mu_A)$ and $k'(\mathbf{x},\mathbf{x}')\doteq k(\mathbf{x},\mathbf{x}')-\mathbf{k}_{\mathbf{x},A}^\top(\mathbf{K}_{AA}+\sigma_{\mathrm{n}}^2\mathbf{I})^{-1}\mathbf{k}_{\mathbf{x}',A}$

- The posterior covariance can only decrease when conditioning on more data, and is independent of $\mathbf{y}_i$.
- GP posterior takes $\mathcal{O}(n^3)$ because of mat. inversion.

Maximizing marginal likelihood: optimizes θ across all realizations of $\mathbf{f}$ (somewhat reg., avoids overfitting): $\hat\theta_{\mathrm{MLE}}\doteq\underset{\theta}{\mathrm{argmax}}\,p(y_{1:n}|\mathbf{x}_{1:n},\theta)=\underset{\theta}{\mathrm{argmax}}\int p(y_{1:n}|\mathbf{x}_{1:n},f,\theta)p(f|\theta)df$

Intuition:

- *Underfit* models: likelihood is mostly small as data cannot be well described; prior is large as there are "fewer" fns. to choose from.
- *Overfit* models: likelihood is large for "some" fns. but small for "most" fns.; prior is small, as probability mass has to be distributed among "more" fns.

Maximizing encourages trading between a large likelihood and large prior, as one product term will be small.

For GP regression: $y_{1:n}|\mathbf{x}_{1:n},\theta\sim\mathcal{N}(\mathbf{0},\mathbf{K}_{f,\theta}+\sigma_{\mathrm{n}}^2\mathbf{I})$, write $\mathbf{K}_{\mathbf{y},\theta}\doteq\mathbf{K}_{f,\theta}+\sigma_{\mathrm{n}}^2\mathbf{I}$, and obtain: $\hat\theta_{\mathrm{MLE}}=\mathrm{argmin}_\theta\underbrace{1/2\cdot\mathbf{y}^\top\mathbf{K}_{\mathbf{y},\theta}^{-1}\mathbf{y}}_{\text{Goodness of fit}}+\underbrace{1/2\cdot\log\det(\mathbf{K}_{\mathbf{y},\theta})}_{\text{"Volume" of model class}}$

The loss can be expressed in closed-form with $\alpha\doteq\mathbf{K}_{\mathbf{y},\theta}^{-1}$: $\frac{\partial}{\partial\theta_j}\log p(y_{1:n}|\mathbf{x}_{1:n},\theta)=\frac{1}{2}\mathrm{tr}((\alpha\alpha^\top-\mathbf{K}_{\mathbf{y},\theta}^{-1})\frac{\partial\mathbf{K}_{\mathbf{y},\theta}}{\partial\theta_j})$

This optimization problem is, in general, non-convex.

- **GPs remain comp. tractable** even though they can model fns. over "infinite-dim." feat. spaces.
- For all $\mathbf{x}\in\mathcal{X}$ and $\mathbf{f}\in\mathcal{H}_k(\mathcal{X})$: $\mathbf{f}(\mathbf{x})=\langle\mathbf{f}(\cdot),k(\mathbf{x},\cdot)\rangle_k$

Given kernel $k:\mathcal{X}\times\mathcal{X}\rightarrow\mathbb{R}$, its **RKHS** is space of fns.: $\mathcal{H}_k(\mathcal{X})\doteq\{f(\cdot)=\sum_{i=1}^n\alpha_ik(\mathbf{x}_i,\cdot)|n\in\mathbb{N},\mathbf{x}_i\in\mathcal{X},\alpha_i\in\mathbb{R}\}$

Inner prod: $\langle f,g\rangle_k\doteq\sum_{i=1}^n\sum_{j=1}^{n'}\alpha_i\alpha_j'k(\mathbf{x}_i,\mathbf{x}'_j)$ where $g(\cdot)=\sum_{j=1}^{n'}\alpha_j'k(\mathbf{x}'_j,\cdot)$. Induces norm $\|\mathbf{f}\|_k=\sqrt{\langle\mathbf{f},\mathbf{f}\rangle_k}$ measuring smoothness/complexity.

- **Representer theorem:** Kernel k, $\lambda>0$, $f\in\mathcal{H}_k(\mathcal{X})$, and train data $\{(\mathbf{x}_i,f(\mathbf{x}_i))\}_{i=1}^n$. Let loss fn. $\mathcal{L}(f(\mathbf{x}_1),\ldots,f(\mathbf{x}_n))\in\mathbb{R}\cup\{\infty\}$ depend on f only through its eval. at train points. Then, any minimizer $\hat f\in\underset{f\in\mathcal{H}_k(\mathcal{X})}{\mathrm{argmin}}\mathcal{L}(f(\mathbf{x}_1),\ldots,f(\mathbf{x}_n))+\lambda\|f\|_k^2$ admits a repr. of form $\hat f(\mathbf{x})=\hat{\mathbf{a}}\mathbf{k}_{\mathbf{x},\{\mathbf{x}_i\}_{i=1}^n}=\sum_{i=1}^n\hat\alpha_ik(\mathbf{x},\mathbf{x}_i)$
- GP MAP is solution of reg. LR problem in RKHS of kernel fn.: $\hat f\doteq\underset{f\in\mathcal{H}_k(\mathcal{X})}{\mathrm{argmin}}-\log(y_{1:n}|\mathbf{x}_{1:n},f)+\frac{1}{2}\|f\|_k^2$
- **GPs remain comp. tractable**

Approximations: GP need to invert mat ($\mathcal{O}(n^3)$).

- **Local method**: When sampling at $\mathbf{x}$ only condition on samples $\mathbf{x}'$, that are close: $|k(\mathbf{x},\mathbf{x}')|\geq\tau$ for some $\tau>0$. **Problem:** τ has to be chosen carefully: if τ is chosen too large, samples become essentially independent. Still expensive if "many" points are close.
- **Kernel approximation**: Construct a low dim. feat. map $\phi:\mathbb{R}^d\rightarrow\mathbb{R}^m$ s.t.: $k(\mathbf{x},\mathbf{x}')\approx\phi(\mathbf{x})^\top\phi(\mathbf{x}')$.
  - Transforms function-space view (GP) back into a tractable weight-space view (BLR, $\mathcal{O}(nm^2+m^3)$).
  - Can be done with **Random Fourier features**: a *stationary* kernel k can be interpreted as fn. in one variable, and has an associated Fourier transform.

Bochner's Theorem A continuous Kernel on $\mathbb{R}^d$ is p.s.d iff its Fourier transform $p(\omega)$ is non-negative.

Rand. feat. map: $z_{\omega,b}(\mathbf{x})\doteq\sqrt{2}\cos(\omega^\top\mathbf{x}+b)$, with $\omega^{(i)}\overset{\mathrm{iid}}{\sim}p$ and $b^{(i)}\overset{\mathrm{iid}}{\sim}\mathrm{Unif}([0,2\pi])$. Inner product $z(x)^\top z(x')$ is unbiased estimator of $k(x-x')$. Error prob. decays exp. in dim. of Fourier feature space m.

- **Inducing pt.**: Summarize data around $U\doteq\{\bar{\mathbf{x}}_1,..\bar{\mathbf{x}}_n\}$. Let $\mathbf{f}\doteq[f(\mathbf{x}_1)..f(\mathbf{x}_n)]^\top$ $f^\star\doteq f(\mathbf{x}^\star)$, $\mathbf{u}\doteq[f(\bar{\mathbf{x}}_1)\ldots f(\bar{\mathbf{x}}_n)]^\top$. Original GP recoverable with marginalization: $p(f^\star|\mathbf{f})=\int_{\mathbb{R}^k}p(f^\star,\mathbf{f}|\mathbf{u})p(\mathbf{u})d\mathbf{u}$. Approx. joint prior, assuming $f^\star$ $\mathbf{f}$ are cond. indep. given $\mathbf{u}\sim\mathcal{N}(\mathbf{0},\mathbf{K}_{UU})$
  Train: $p(\mathbf{f}|\mathbf{u})\sim\mathcal{N}(\mathbf{f};\mathbf{K}_{AU}\mathbf{K}_{UU}^{-1}\mathbf{u},\mathbf{K}_{AA}-\mathbf{Q}_{AA})$
  Test: $p(f^\star|\mathbf{u})\sim\mathcal{N}(f^\star;\mathbf{K}_{\star U}\mathbf{K}_{UU}^{-1}\mathbf{u},\mathbf{K}_{\star\star}-\mathbf{Q}_{\star\star})$ w. $\mathbf{Q}_{ab}\doteq\mathbf{K}_{aU}\mathbf{K}_{UU}^{-1}\mathbf{K}_{Ub}$. $\mathbf{K}_{AA}$ represents the prior covar. and $\mathbf{Q}_{AA}$ represents covar. from inducing pts.
  Covar. mat. comp. is expensive; need to approx.:
  - **Subset of regressors (SoR)**: Forgets about all var. and covar. $q_{\mathrm{SoR}}(\mathbf{f}|\mathbf{u})\doteq\mathcal{N}(\mathbf{f};\mathbf{K}_{AU}\mathbf{K}_{UU}^{-1}\mathbf{u},\mathbf{0})$
    $q_{\mathrm{SoR}}(f^\star|\mathbf{u})\doteq\mathcal{N}(f^\star;\mathbf{K}_{\star U}\mathbf{K}_{UU}^{-1}\mathbf{u},0)$
  - **Fully independent training conditional (FITC)**: Keeps track of variances but forgets about covariance $q_{\mathrm{FITC}}(\mathbf{f}|\mathbf{u})\doteq\mathcal{N}(\mathbf{f};\mathbf{K}_{AU}\mathbf{K}_{UU}^{-1}\mathbf{u},\mathrm{diag}\{\mathbf{K}_{AA}-\mathbf{Q}_{AA}\})$
    $q_{\mathrm{FITC}}(f^\star|\mathbf{u})\doteq\mathcal{N}(f^\star;\mathbf{K}_{\star U}\mathbf{K}_{UU}^{-1}\mathbf{u},\mathrm{diag}\{\mathbf{K}_{\star\star}-\mathbf{Q}_{\star\star}\})$

Comp. cost SoR/FITC is dom. by mat. inv. of $\mathbf{K}_{UU}$, so cubic in num. inducing pts. and linear in data pts.

4 Variational Inference

Idea: approximate true posterior distribution with a simpler posterior that is easy to sample: $p(\theta|\mathbf{x}_{1:n},y_{1:n})=\frac{1}{Z}p(\theta,y_{1:n}|\mathbf{x}_{1:n})\approx q(\theta|\lambda)\doteq q_\lambda(\theta)$, where λ are params. of the **variational posterior** q_λ.

Laplace approx.: find Gaussian approx. (2nd-order Taylor) of posterior: $q(\theta)\doteq\mathcal{N}(\theta;\hat\theta,\mathbf{\Lambda}^{-1})\propto\exp(\hat\psi(\theta))$, with $\hat\theta$ the mode (i.e. MAP estimate) and with $\mathbf{H}$ the Hessian: $\mathbf{\Lambda}\doteq-\mathbf{H}_\psi(\hat\theta)=-\mathbf{H}_\theta\log p(\theta|\mathbf{x}_{1:n},y_{1:n})\big|_{\theta=\hat\theta}$. Perform inference using the variations approximation: $p(y^\star|\mathbf{x}^\star,\mathbf{x}_{1:n},y_{1:n})\approx\int p(y^\star|\mathbf{x}^\star,\theta)q_\lambda(\theta)d\theta=\mathbb{E}_{\theta\sim q_\lambda}[p(y^\star|\mathbf{x}^\star,\theta)]$

- Matches shape of true posterior around its mode but may not represent it accurately elsewhere.
- Leads to extremely overconfident predictions, often unsuitable for approximate probabilistic inference.
- Preserves MAP point estimate as its mean.
- Laplace approximation of a Gaussian is exact.
- **For log. BLR:** $\mathbf{\Lambda}=\mathbf{X}^\top\mathrm{diag}\{\pi_i(1-\pi_i)\}\mathbf{X}+\sigma_{\mathrm{p}}^{-2}\mathbf{I}$ (indep. of norm.const.) $p(y^\star|\mathbf{x}^\star,\mathbf{x}_{1:n},y_{1:n})\approx\int\sigma(y^\star\mathbf{w}^\top\mathbf{x})\mathcal{N}(\mathbf{w};\hat{\mathbf{w}},\Lambda^{-1})d\mathbf{w}=\int\sigma(y^\star f)\mathcal{N}(f;\hat{\mathbf{w}}^T\mathbf{x}^\star,\mathbf{x}^{\star T}\Lambda^{-1}\mathbf{x}^\star)df$

Jensen's inequality: For convex fn. g we have: $g(\mathbb{E}[X])\leq\mathbb{E}[g(X)]$; if h concave: $h(\mathbb{E}[X])\geq\mathbb{E}[h(X)]$

Surprise of an event with probability u: $\mathrm{S}[u]\doteq-\log u$. $\mathrm{S}[u]$ is convex in u. For a discrete RV X: $\mathrm{S}[p(x)]\geq0$. Axiomatic characterization up to pos. const. factor:

- $\mathrm{S}[u]>\mathrm{S}[v]\Longrightarrow u<v$ (anti-monoton) • S continuous
- $\mathrm{S}[uv]=\mathrm{S}[u]+\mathrm{S}[v]$ for independent events

The **cross-entropy** of q relative to p is: $\mathrm{H}[p\|q]\doteq\mathbb{E}_{x\sim p}[\mathrm{S}[q(x)]]=\mathbb{E}_{x\sim p}[-\log q(x)]$.

Entropy of distr. p is avg. surprise of samples from p: $\mathrm{H}[p]\doteq\mathbb{E}_{x\sim p}[\mathrm{S}[p(x)]]=\mathbb{E}_{x\sim p}[-\log p(x)]$. Can be negative but if p is discrete then $\mathrm{H}[p]\geq0$.

KL-div.: measures additional expected surprise when observing samples from p that is due to assuming (wrong) q and which is not inherent in p already. $\mathrm{KL}(p\|q)\doteq\mathrm{H}[p\|q]-\mathrm{H}[p]=\mathbb{E}_{\theta\sim p}[\log\frac{p(\theta)}{q(\theta)}]$

- $\mathrm{KL}(p\|q)\geq0$; $\mathrm{KL}(p\|q)=0$ iff $p=q$ almost surely
- There exist distr. p and q s.t. $\mathrm{KL}(p\|q)\neq\mathrm{KL}(q\|p)$

Note that: $\mathrm{H}[p\|q]=\mathrm{H}[p]+\mathrm{KL}(p\|q)\geq\mathrm{H}[p]$.

- $\mathrm{KL}(\mathrm{Bern}(p)\|\mathrm{Bern}(q))=p\log\frac{p}{q}+(1-p)\log\frac{(1-p)}{(1-q)}$
- For $p\doteq\mathcal{N}(\mu_p,\mathbf{\Sigma}_p)$ and $q\doteq\mathcal{N}(\mu_q,\mathbf{\Sigma}_q)$: $\mathrm{KL}(p\|q)=\frac{1}{2}\left(\mathrm{tr}(\mathbf{\Sigma}_q^{-1}\mathbf{\Sigma}_p)+(\mu_p-\mu_q)^\top\mathbf{\Sigma}_q^{-1}(\mu_p-\mu_q)-d+\log\frac{\det(\mathbf{\Sigma}_q)}{\det(\mathbf{\Sigma}_p)}\right)$

- **Forward KL**: $q_1^\star\doteq\mathrm{argmin}_{q\in\mathcal{Q}}\mathrm{KL}(p\|q)$(mode avg., more conservative, yields more "desired" approx.)
- **Reverse KL**: $q_2^\star\doteq\mathrm{argmin}_{q\in\mathcal{Q}}\mathrm{KL}(q\|p)$ (greedily mode seeking, underestimate var., overconfident preds.)

Evidence lower bound (ELBO), for given data $\mathcal{D}_n$: $L(q,p;\mathcal{D}_n)=\underbrace{\log p(y_{1:n}|\mathbf{x}_{1:n})}_{\text{const}}-\mathrm{KL}(q\|p(\cdot|\mathbf{x}_{1:n},y_{1:n}))$ $=\underbrace{\mathbb{E}_{\theta\sim q}[\log p(y_{1:n}|\mathbf{x}_{1:n},\theta)]}_{\text{log-likelihood}}-\underbrace{\mathrm{KL}(q\|p(\cdot))}_{\text{proximity to prior}}$

Max the ELBO coincides with min. reverse-KL. Since KL-div. is non-negative: $\log p(y_{1:n}|\mathbf{x}_{1:n})\geq L(q,p;\mathcal{D}_n)$

- Max. ELBO selects a var. distr. q that is close to prior $p(\cdot)$ while also max. avg. data likelihood $p(y_{1:n}|\mathbf{x}_{1:n},\theta)$ for $\theta\sim q$. Contrast to MAP, which picks single mode θ that max. the likelihood and proximity to the prior.
- ELBO gradient is gen. **intractable** (use rep. trick).

Repearm. trick: Let $\epsilon\sim\phi$ be indep. of λ, $\mathbf{g}:\mathbb{R}^d\rightarrow\mathbb{R}^d$ be a diff. and inv. fn, $\theta\doteq\mathbf{g}(\epsilon;\lambda)$, and $\mathbf{f}$

- Goal: approx. true posterior $p(\theta|\mathcal{D})$ with simpler variational distr. q_λ typically family of indep. Gaussians.
- Achieved by max. ELBO with SGD and rep. trick.
- We can approx. the predictive distr. by sampling from the variational posterior $p(y^\star|\mathbf{x}^\star,\mathbf{x}_{1:n},\mathbf{y}_{1:n})\approx \mathbb{E}_{\theta\sim q_\lambda}[p(y^\star|\mathbf{x}^\star,\theta)]\approx\frac{1}{m}\sum_{i=1}^m p(y^\star|\mathbf{x}^\star,\theta^{(i)})$.
- VI in BNNs can be seen as avg. preds. of multiple NNs drawn acc. to the variational posterior q_λ.
- Using Monte Carlo samples estimate mean and var.:
$\mathbb{E}[y^\star|\mathbf{x}^\star,\mathbf{x}_{1:n},\mathbf{y}_{1:n}]\approx\frac{1}{m}\sum_{i=1}^m\mu(\mathbf{x}^\star;\theta^{(i)})$
$\mathrm{Var}[y^\star|\mathbf{x}^\star,\mathbf{x}_{1:n},\mathbf{y}_{1:n}]\approx\mathbb{E}_\theta[\mathrm{Var}_{y^\star}[y^\star|\mathbf{x}^\star,\theta]]+\mathrm{Var}_\theta[\mathbb{E}_{y^\star}[y^\star|\mathbf{x}^\star,\theta]]$
$\approx\mathbb{E}_\theta[\sigma^2(\mathbf{x}^\star;\theta)]+\mathrm{Var}_\theta[\mu(\mathbf{x}^\star;\theta)]$
$\approx\underbrace{\frac{1}{m}\sum_{i=1}^m\sigma^2(\mathbf{x}^\star;\theta^{(i)})}_{\text{aleatoric}}+\underbrace{\frac{1}{m-1}\sum_{i=1}^m(\mu(\mathbf{x}^\star;\theta^{*(i)})-\bar\mu(\mathbf{x}^\star))^2}_{\text{epistemic}}$

- **Dropout/Dropconnect** random select/omits vertices/edges of comp. graph. For valid interpretation of this as VI, also need to perform during inference.
- Dropout masks will overlap, making preds. highly corr., leading to underest. of epistemic uc. **Masksembles** mitigate by choosing fixed set of pre-defined masks.
- **Probabilistic ensembles:** learn m different NN over random chosen subsets of train data for each network.

Evidence of val. set; How well model desc. val. set?: $\log p(y_{1:M}^{\text{val}}|\mathbf{x}_{1:m}^{\text{val}},\mathbf{x}_{1:n}^{\text{train}},y_{1:n}^{\text{train}})\gtrsim\frac{1}{k}\sum_{j=1}^k\sum_{i=1}^m\log p(y_i^{\text{val}}|\mathbf{x}_i^{\text{val}},\theta^{(j)})$

- **Frequency:** Proportion of samples in bin m that belong to 1: $\mathrm{freq}(B_m)\doteq\frac{1}{|B_m|}\sum_{i\in B_m}\mathbf{1}\{\mathcal{Y}_i=1\}$
- **Confidence:** Avg. conf. of samples in bin m belonging to 1: $\mathrm{conf}(B_m)\doteq\frac{1}{|B_m|}\sum_{i\in B_m}\mathbb{P}\{\mathcal{Y}_i=1|\mathbf{x}_i\}$

A model is **well-calibrated** if its confidence coincides with its acc. across many preds.: $\mathrm{freq}(B_m)\approx\mathrm{conf}(B_m)$

- **ECE:** $\ell_{\mathrm{ECE}}\doteq\sum_{m=1}^M\frac{|B_m|}{n}|\mathrm{freq}(B_m)-\mathrm{conf}(B_m)|$
- **MCE:** Max instead of sum

6 Active Learning

Cond. entropy: $\mathrm{H}[\mathbf{X}|\mathbf{Y}]\doteq\mathbb{E}_{\mathbf{y}\sim p(\mathbf{y})}[\mathrm{H}[\mathbf{X}|\mathbf{Y}=\mathbf{y}]]$
$=\mathbb{E}_{(\mathbf{x},\mathbf{y})\sim p(\mathbf{x},\mathbf{y})}[-\log p(\mathbf{x}|\mathbf{y})]$

Joint entropy: $\mathrm{H}[\mathbf{X},\mathbf{Y}]\doteq\mathbb{E}_{(\mathbf{x},\mathbf{y})\sim p(\mathbf{x},\mathbf{y})}[-\log p(\mathbf{x},\mathbf{y})]$
- $\mathrm{H}[\mathbf{X}|\mathbf{Y}]\neq\mathrm{H}[\mathbf{Y}|\mathbf{X}]$ in general; but $\mathrm{H}[\mathbf{X},\mathbf{Y}]=\mathrm{H}[\mathbf{Y},\mathbf{X}]$
- $\mathrm{H}[\mathbf{X},\mathbf{Y}]=\mathrm{H}[\mathbf{Y}]+\mathrm{H}[\mathbf{X}|\mathbf{Y}]=\mathrm{H}[\mathbf{X}]+\mathrm{H}[\mathbf{Y}|\mathbf{X}]$
- $\mathrm{H}[\mathbf{X}|\mathbf{Y}]=\mathrm{H}[\mathbf{Y}|\mathbf{X}]+\mathrm{H}[\mathbf{X}]-\mathrm{H}[\mathbf{Y}]$ (Bayes Rule)
- $\mathrm{H}[\mathbf{X}|\mathbf{Y}]\leq\mathrm{H}[\mathbf{X}]$ (Gibbs; Information never hurts) $\Leftrightarrow 0\leq\mathrm{H}[\mathbf{X}]-\mathrm{H}[\mathbf{X}|\mathbf{Y}]=\mathrm{I}(\mathbf{X};\mathbf{Y})$

Mutual info: $\mathrm{I}(\mathbf{X};\mathbf{Y})\doteq\mathrm{H}[\mathbf{X}]+\mathrm{H}[\mathbf{Y}]-\mathrm{H}[\mathbf{X},\mathbf{Y}]$
- $\mathrm{I}(\mathbf{X};\mathbf{Y})=\mathrm{I}(\mathbf{Y};\mathbf{X})=\mathbb{E}_{\mathbf{y}\sim p}[\mathrm{KL}(p(\mathbf{x}|\mathbf{y})\|p(\mathbf{x}))]$

Cond. mutual info:
- $\mathrm{I}(\mathbf{X};\mathbf{Y}\mid\mathbf{Z})=\mathrm{H}[\mathbf{X}|\mathbf{Z}]-\mathrm{H}[\mathbf{X}|\mathbf{Y},\mathbf{Z}]$
$=\mathrm{H}[\mathbf{X},\mathbf{Z}]+\mathrm{H}[\mathbf{Y},\mathbf{Z}]-\mathrm{H}[\mathbf{Z}]-\mathrm{H}[\mathbf{X},\mathbf{Y},\mathbf{Z}]$
$=\mathrm{I}(\mathbf{X};\mathbf{Y},\mathbf{Z})-\mathrm{I}(\mathbf{X};\mathbf{Z})$
- $\mathrm{I}(\mathbf{X};\mathbf{Y}|\mathbf{Z})=\mathrm{I}(\mathbf{Y};\mathbf{X}|\mathbf{Z})$
- $\mathrm{I}(\mathbf{X};\mathbf{Y};\mathbf{Z})\doteq\mathrm{I}(\mathbf{X};\mathbf{Y})-\mathrm{I}(\mathbf{X};\mathbf{Y}|\mathbf{Z})$, so the "information never hurts" principle does not hold for MI. Information about Z may reduce the MI between $\mathbf{X}$ and $\mathbf{Y}$

- Given (discrete) fn. $F:\mathcal{P}(\mathcal{X})\to\mathbb{R}$, the **marginal gain** of $\mathbf{x}\in\mathcal{X}$ given $A\subseteq\mathcal{X}$ is: $\Delta_F(\mathbf{x}|A)\doteq F(A\cup\{\mathbf{x}\})-F(A)$.
- The fn. is **submodular** iff for any $\mathbf{x}\in\mathcal{X}$ and any $A\subseteq B\subseteq\mathcal{X}$: $F(A\cup\{\mathbf{x}\})-F(A)\geq F(B\cup\{\mathbf{x}\})-F(B)$ or equally $\Delta_F(\mathbf{x}|A)\geq\Delta_F(\mathbf{x}|B)$. Submodularity can be interpreted as notion of "concavity" for discrete fns.
- It is called **monotone** if $F(A)\leq F(B)$.

Maximization objective: monotone submodular function: $I(S)\doteq\mathrm{I}(\mathbf{f}_S;\mathbf{y}_S)=\mathrm{H}[\mathbf{f}_S]-\mathrm{H}[\mathbf{f}_S|\mathbf{y}_S]$. $\mathrm{H}[\mathbf{f}_S]$: uc in $\mathbf{f}_S$ before observing $\mathbf{y}_S$. $\mathrm{H}[\mathbf{f}_S|\mathbf{y}_S]$ uc in $\mathbf{f}_S$ after observing $\mathbf{y}_S$. Max. MI is in general NP-hard.

- **Greedy**: Pick $\mathbf{x}_1$ through $\mathbf{x}_n$ individually by greedily finding the location with the maximal MI, this provides a $(1-1/e)$-approximation of the optimum.
- **Uncertainty sampling**: Have already picked $S_t=\{\mathbf{x}_1,\dots,\mathbf{x}_t\}$; Solve the following: $\mathbf{x}_{t+1}\doteq\mathrm{argmax}_{\mathbf{x}\in\mathcal{X}}\Delta_I(\mathbf{x}|S_t)=\mathrm{argmax}_{\mathbf{x}\in\mathcal{X}}\mathrm{I}(f_\mathbf{x};y_\mathbf{x}|\mathbf{y}_{S_t})$. Doesn't work with heteroscedastic noise: large aleatoric uc may dominate epistemic uc. In classification corresponds to selecting label that max. entropy of predicted label: $\mathbf{x}_{t+1}\doteq\mathrm{argmax}_{\mathbf{x}\in\mathcal{X}}\mathrm{H}[y_\mathbf{x}|\mathbf{x}_{1:t},y_{1:t}]$.

Bayesian active learning by disagreement (BALD): Identifies points $\mathbf{x}$ where models *disagree* about label $y_\mathbf{x}$ (each model is *confident* but predict different labels): $\mathbf{x}_{t+1}\doteq\mathrm{argmax}_{\mathbf{x}\in\mathcal{X}}\mathrm{I}(\theta;y_\mathbf{x}|\mathbf{x}_{1:t},y_{1:t})=\mathrm{argmax}_{\mathbf{x}\in\mathcal{X}}\mathrm{H}[y_\mathbf{x}|\mathbf{x}_{1:t},y_{1:t}]-\mathbb{E}_{\theta|\mathbf{x}_{1:t},y_{1:t}}\mathrm{H}[y_\mathbf{x}|\theta]$

- **Inductive learn.** extract general rules from data. Typically, we can directly observe $f(\mathbf{x})$ at any $\mathbf{x}$.
- **Transductive learn.** cannot directly observe target $f(\mathbf{x}^\star)$. Requires generalization from other locations.

7 Bayesian Optimization

Cum. regret: $R_T\doteq\sum_{t=1}^T(\max_\mathbf{x}f^\star(\mathbf{x})-f^\star(\mathbf{x}_t))$
Instant regret: $(\max_\mathbf{x}f^\star(\mathbf{x})-f^\star(\mathbf{x}_t))$

Goal: Achieve **sublinear regret**: $\lim_{T\to\infty}R_T/T=0$ (requires balancing exploration and exploitation).

Algorithm 9.2: Bayesian optimization (with GPs)

initialize $f\sim\mathcal{GP}(\mu_0,k_0)$
for $t=1$ **to** T **do**
  choose $\mathbf{x}_t=\arg\max_{x\in\mathcal{X}}F(\mathbf{x};\mu_{t-1},k_{t-1})$
  observe $y_t=f(\mathbf{x}_t)+\epsilon_t$
  perform a probabilistic update to obtain μ_t and k_t

- Common to use an **acquisition fn.** to greedily pick the next point to sample based on the current model.
- **UCB**: $\mathbf{x}_{t+1}\doteq\mathrm{argmax}_{\mathbf{x}\in\mathcal{X}}\mu_t(\mathbf{x})+\beta_{t+1}\sigma_t(\mathbf{x})$, where $\sigma_t(\mathbf{x})\doteq\sqrt{k_t(\mathbf{x},\mathbf{x})}$. Generally non-convex. If $\beta_t=0$: purely exploitative; if $\beta_t\to\infty$: recovers uc sampling. UCB curves must pass through datapoints since $\sigma=0$.
- **PI**: likelihood that $\mathbf{x}$ will improve current best by at least ξ. Comp. efficient. Can be overly "greedy" by taking pts. with high prob. of small improvement.
- **EI**: exp. of improvement $\mathrm{I}(x)=\max(0,f(\mathbf{x})-f(\mathbf{x}^\star))$ over current best $f(\mathbf{x}^\star)$. Closed-form solution for GPs. Balances pts. with high pred. value and those with high uc. Problem: Vanishing gradients (use log to help).

When choosing β_t appropriately: $R_T=\mathcal{O}(\sqrt{T\gamma_T})$, with $\gamma_T\doteq\max_{S\subset\mathcal{X},|S|=T}\mathrm{I}(\mathbf{f}_S;\mathbf{y}_S)=\max_{S\subset\mathcal{X},|S|=T}\frac{1}{2}\log\det(\mathbf{I}+\sigma_\mathrm{n}^{-2}\mathbf{K}_{SS})$ is the maximum information gain after T rounds.
- Linear: $\gamma_T=\mathcal{O}(d\log T)$
- Gaussian: $\gamma_T=\mathcal{O}((\log T)^{d+1})$
- Matérn for $\nu>\frac{1}{2}$: $\gamma_T=\mathcal{O}(T^{\frac{d}{2\nu+d}}(\log T)^{\frac{2\nu}{2\nu+d}})$

Thompson Sampling: At time $t+1$, we sample a fn. $\tilde f_{t+1}\sim p(\cdot|\mathbf{x}_{1:t},y_{1:t})$ from our posterior distr. Then, we simply max. $\tilde f_{t+1}$, $\mathbf{x}_{t+1}\doteq\mathrm{argmax}_{\mathbf{x}\in\mathcal{X}}\tilde f_{t+1}(\mathbf{x})$.

8 Diffusion generative models

Let $\beta_t\in(0,1]$, $\bar\alpha_t=\prod_{s=1}^t\alpha_s$, and $\alpha_s=1-\beta_s$. Typically, β_t is monotonically increases, which implies that $\bar\alpha_t\to0$ and thus $x_T\to\mathcal{N}(0,\mathbf{I})$ for $T\to\infty$.

1. **Forward process:** Transform data points into (Gaussian) noise by using a fixed noising MC q:
$q(x_{1:T}|x_0)=\prod_{t=1}^Tq(x_t|x_{t-1})$ (MC-chain)
$q(x_t|x_{t-1})=\mathcal{N}(x_t;\sqrt{1-\beta_t}x_{t-1},\beta_t\mathbf{I})$
$q(x_t|x_0)=\mathcal{N}(x_t;\sqrt{\bar\alpha_t}x_0,(1-\bar\alpha_t)\mathbf{I})$
2. **Backward process:** Learn a denoising MC p matching the reversed forward process.
$p_\lambda(x_{t-1}|x_t)=\mathcal{N}(x_{t-1};\mu_\lambda(x_t,t),\mathbf{\Sigma}_\lambda(x_t,t))$
$p_\lambda(x_{0:T})=p_\lambda(x_T)\prod_{t=1}^Tp_\lambda(x_{t-1}|x_t)$
$p_\lambda(x_0)=\int p_\lambda(x_{0:T})dx_{1:T}$ where $x_{1:T}$ latent vars.
3. **Generation:** Now generate novel data points by simulating the learned denoising MC p. Sample seq.: (1) $x_1\sim p(X_1)$, (2) $x_2\sim p(X_2|X_1=x_1)$, ...

Use VI. ELBO to get intractable $p_\lambda(x_0)$:
$\log p_\lambda(x_0)\geq\log p_\lambda(x_0)-\mathcal{D}_{\mathrm{KL}}(q(\cdot|x_0)\|p_\lambda(\cdot|x_0))$
$=\mathrm{const}+\mathbb{E}_\tau[-\underbrace{\sum_{t=2}^T\mathcal{D}_{\mathrm{KL}}(q(\cdot|x_t,x_0)\|p_\lambda(\cdot|x_t))}_{L_t}+\underbrace{\log p_\lambda(x_0|x_1)}_{L_1}]$
w. $\tau\doteq x_{1:T}\sim q(\cdot|x_0)$. Assume const. var. schedule $\Sigma_t=\sigma_t^2\mathbf{I}$ and optimize **closed-forms** via **stochastic VI**:
$L_t=1/2\sigma_t^2\|\mu_t'(x_t,x_0)-\mu_\lambda(x_t,t)\|_2^2+\mathrm{const}$
Closed-form $\mu_t'(x_t,x_0)$ (can be highly non-stationary):
$\mu_t'(x_t,x_0)=\frac{\sqrt{\bar\alpha_t}\beta_t}{1-\bar\alpha_t}x_0+\frac{\sqrt{\alpha_t}(1-\bar\alpha_{t-1})}{(1-\bar\alpha_t)}x_t$, use rep. trick:
$x_t(x_0,\epsilon)=\sqrt{\bar\alpha_t}x_0+\sqrt{1-\bar\alpha_t}\epsilon$ and learn sample noise.
Sample: $x_{t-1}=\frac{1}{\sqrt{\alpha_t}}(x_t-\frac{1-\alpha_t}{\sqrt{1-\bar\alpha_t}}\epsilon_\lambda(x_t,t))+\sigma_tz$

9 Markov Decision Processes (MDPs)

Discounted payoff: $G_t\doteq\sum_{m=0}^\infty\gamma^mR_{t+m}$

- **State value fn.:** $v_t^\pi\doteq\mathbb{E}_\pi[G_t|X_t=x,A_t=a]$ measures avg. discounted payoff from time t starting from $x\in X$.
- **State-action value fn. (Q-fn.):** $q_t^\pi(x,a)\doteq\mathbb{E}_\pi[G_t|X_t=x,A_t=a]=r(x,a)+\gamma\sum_{x'\in X}p(x'|x,a)\cdot v_{t+1}^\pi(x')$ measures avg. discounted payoff from time t starting from $x\in X$ and with playing action $a\in A$.

Bellman Expectation Equation:
- $v^\pi(x)=r(x,\pi(x))+\gamma\mathbb{E}_{x'|x,\pi(x)}[v^\pi(x')]$
- If stochastic policy: $v^\pi(x)=\mathbb{E}_{a\sim\pi(x)}[q^\pi(x,a)]$
$q^\pi(x,a)=r(x,a)+\gamma\mathbb{E}_{x'|x,a}\mathbb{E}_{a'\sim\pi(x')}[q^\pi(x',a')]$
- For deterministic: $v^\pi(x)=q^\pi(x,\pi(x))$.

Can be used to find v^π given policy π, by solving linear system of eq. in cubic time in size of state space. Can also be solved using fixed pt. iter: $\mathbf{B}^\pi\mathbf{v}\doteq\mathbf{r}^\pi+\gamma\mathbf{P}^\pi\mathbf{v}$.
$\|\mathbf{v}_t^\pi-\mathbf{v}^\pi\|_\infty=\|\mathbf{B}^\pi\mathbf{v}_{t-1}^\pi-\mathbf{B}^\pi\mathbf{v}^\pi\|_\infty\leq\gamma\|\mathbf{v}_{t-1}^\pi-\mathbf{v}^\pi\|_\infty\leq\gamma\|\mathbf{v}_0^\pi-\mathbf{v}^\pi\|_\infty$

Bellman's theorem: A policy $\pi^\star$ is optimal iff it's greedy w.r.t its own value fn. In other words, $\pi^\star$ is optimal iff $\pi^\star(x)$ is a distr. over set $\mathrm{argmax}_{a\in A}q^\star(x,a)$.

Optimal $v^\star$ and $q^\star$ are fixed-pt. of Bellman update.

Bellman optimality equations:
$v^\star(x)=\max_{a\in A}r(x,a)+\gamma\mathbb{E}_{x'|x,a}[v^\star(x')]$
$q^\star(x,a)=r(x,a)+\gamma\mathbb{E}_{x'|x,a}[\max_{a\in A}q^\star(x',a')]$

- If for every state there is a unique action that max. the q-fn., $\pi^\star$ is deter. and unique. $\pi^\star(x)=\mathrm{argmax}_{a\in A}q^\star(x,a)$.
- For finite MDPs, PI converges to $\pi^\star$ in poly. num. of iter. Each step takes cubic comp. in the num. of states.

Algorithm 10.14: Policy iteration

initialize π (arbitrarily)
repeat
  compute v^π
  compute π_{v^π}
  $\pi\leftarrow\pi_{v^\pi}$
until converged

Monotonic improvement of PI:
- $v^{\pi_{t+1}}(x)\geq v^{\pi_t}(x)$ for all $x\in X$
- $v^{\pi_{t+1}}(x)>v^{\pi_t}(x)$ for at least one $x\in X$, unless $v^{\pi_t}\equiv v^\star$

Algorithm 10.17: Value iteration

initialize $v(x)\leftarrow\max_{a\in A}r(x,a)$ for each $x\in X$
for $t=1$ **to** ∞ **do**
  $v(x)\leftarrow(\boldsymbol{B}^\star v)(x)=\max_{a\in A}q(x,a)$ for each $x\in X$
choose π_v

- VI converges to an optimal policy, as $v^\star$ and $q^\star$ are a fixed-points of the Bellman update $\mathbf{B}^\star$.
- For any $\epsilon>0$, VI converges to an ϵ-optimal solution in poly time. However, unlike PI, VI does not generally reach the exact optimum in a finite num. of iter.
- VI can be viewed from dynamic prog. perspective

POMDP: Markov process, with **observations** Y, and **observation probs.** $o(y|x)\doteq\mathbb{P}(Y_t=y|X_t=x)$. Hard to solve in gen., can conv. to MDP with larger state space.

10 Tabular Reinforcement Learning

Markovian property of the underlying MDP:
$X_{t=1}\perp X_{t'+1}|X_t,X_{t'+1},A_t,A_{t'}\quad R_t\perp X_{t'}|R_t,X_{t'+1},A_t,A_{t'}$

Robbins-Montro (RM) conditions: for a sequence $(\alpha)_{t\in\mathbb{N}_0}$ if: $\alpha_t\geq0$, $\sum_{t=0}^\infty\alpha_t=\infty$, $\sum_{t=0}^\infty\alpha_t^2<\infty$.

Bootstrapping: approx. a true quantity by using an empirical quantity, which itself is constructed using samples from the true quantity that is to be approx.

For model-based approaches MLE yields:
$\hat p(x'|x,a)=\frac{N(x'|x,a)}{N(a|x)}$ and $\hat r(x,a)=\frac{1}{N(a|x)}\sum_{t=0,x_t=x,a_t=a}^\infty r_t$
Both unbiased as they correspond to a sample mean.
- $N(x'|x,a)$ num. trans. from x to x' when play a
- $N(a|x)$ nums trans. from x and play a.

Greedy in the limit with inf. exploration (GLIE):
1. All state-action pairs are explored infinitely many times: $\lim_{t\to\infty}N_t(x,a)=\infty$
2. The policy converges to a greedy policy: $\lim_{t\to\infty}\pi_t(a|x)=\mathbf{1}\{a=\mathrm{argmax}_{a'\in A}Q_t^\star(x,a')\}$

ϵ-greedy: • Ignores all past experience. • Will eventually converge. • Will eventually converge. GLIE with prob. 1 if $(\epsilon_t)_{t\in\mathbb{N}_0}$ satisfies the RM conditions (e.g., $\epsilon_t=1/t$).

Softmax/Boltzmann exploration: alt. to ϵ-greedy $\pi_\lambda(a|x)\propto\exp(\frac{1}{\lambda}Q^\star(x,a))$ (Gibbs). For $\lambda\to0$ greedily max. Q-fn. For $\lambda\to\infty$ uniform rand. exploration.

Algorithm 11.6: $R_{\max}$ algorithm

add the fairy-tale state x^* to the Markov decision process
set $\hat r(x,a)=R_{\max}$ for all $x\in X$ and $a\in A$
set $\hat p(x^*\mid x,a)=1$ for all $x\in X$ and $a\in A$
compute the optimal policy $\hat\pi$ for $\hat r$ and $\hat p$
for $t=0$ **to** ∞ **do**
  execute policy $\hat\pi$ (for some number of steps)
  for each visited state-action pair (x,a), update $\hat r(x,a)$
  estimate transition probabilities $\hat p(x'\mid x,a)$
  after observing "enough" transitions and rewards, recompute the optimal policy $\hat\pi$ according the current model $\hat p$ and $\hat r$.

On-policy, Model-based

- Optimism in the face of uc. Init. with max reward.
- Every T steps, with high prob., either obtains near-optimal rew.; or visits one unknown state-action pair.
- With prob. at least $1-\delta$, $R_{\max}$ reaches ϵ-optimal π in poly. num. steps $|X|$, $|A|$, T, $1/\epsilon$, $1/\delta$, and $R_{\max}$.

Algorithm 11.9: Temporal-difference (TD) learning

initialize V^π arbitrarily (e.g., as $\mathbf{0}$)
for $t=0$ **to** ∞ **do**
  follow policy π to obtain the transition (x,a,r,x')
  $V^\pi(x)\leftarrow(1-\alpha_t)V^\pi(x)+\alpha_t(r+\gamma V^\pi(x'))$

On-policy, Model-free

- If α_t satisfies RM conditions and all state-action pairs are chosen inf. often, then V^π conv. to v^π w. prob 1.
- For estimates V^π to converge true v^π, the transitions that are used for the estimation must follow policy π.

SARSA: Same as TD but estimate Q with update: $Q^\pi(x,a)\leftarrow(1-\alpha_t)Q^\pi(x,a)+\alpha_t(r+\gamma Q^\pi(x',a'))$
Same convergence guarantees as TD. On-policy, Model-free

Algorithm 11.12: Q-learning

initialize $Q^\star(x,a)$ arbitrarily (e.g., as $\mathbf{0}$)
for $t=0$ **to** ∞ **do**
  observe the transition (x,a,r,x')
  $Q^\star(x,a)\leftarrow(1-\alpha_t)Q^\star(x,a)+\alpha_t(r+\gamma\max_{a'\in A}Q^\star(x',a'))$

Off-policy, Model-free

- If α_t satisfies RM cond. and all state-action pair are visited inf. often, then $Q^\star$ conv. to $q^\star$ with prob. 1.
- With prob. at least $1-\delta$, conv. to ϵ-optimal policy in num. steps poly. in $\log|X|$, $\log|A|$, $1/\epsilon$ and $\log1/\delta$.

Optimistic Q-learning: Similar to $R_{\max}$ Init.
$Q^\star(x,a)=V_{\max}\prod_{t=1}^{T_{\text{init}}}(1-\alpha_t)^{-1}$ w.
$V_{\max}\doteq\frac{R_{\max}}{1-\gamma}\geq\max q^\star(x,a)$. With prob. at least $1-\delta$, ϵ-optimal π after num. steps poly. in $|X|$, $|A|$, $\frac{1}{\epsilon}$, $\log\frac{1}{\delta}$, and $R_{\max}$ where T_{init} is upper bounded by a poly. in same coeff. If T_{init} large enough, conv. quickly to $\pi^\star$.

11 Model-free Reinforcement Learning

- Time req. to compute value fn. for every state-action pair exactly grows poly. in size of state-action space.
- Can view TD-/Q-learning as SGD on the squared loss: $\ell(\theta;x,r,x')\doteq\frac{1}{2}(r+\gamma\theta^{\text{old}}(x')-\theta(x))^2$ and learn param. approx. of $V(\mathbf{x};\theta)$ or $Q(\mathbf{x},\mathbf{a};\theta)$ using Monte Carlo est. and bootstrapping.

Model-free (TD-/Q-learn.) are usually **sample ineff.**:
- Bootstrapping leads to "(initially) incorrect" and "unstable" targets of the optimization
- Monte Carlo est. with single sample leads to large var.

Q-learning with fn. approx.: (1) Observe $\mathbf{x}'$ and r from picking a in $\mathbf{x}$. (2) Update $\theta\leftarrow\theta+\alpha_t\delta_\mathrm{B}\phi(\mathbf{x},\mathbf{a})$, where $\delta_\mathrm{B}\doteq r+\gamma\max_{\mathbf{a}'\in A}Q^\star(\mathbf{x}',\mathbf{a}';\theta^{\text{old}})-Q^\star(\mathbf{x},\mathbf{a};\theta)$.

- In the tabular setting, this is identical to Q-learning
- Converges to the true Q-function $q^\star$.

"Tricks of the trade" to improve SGD:
- **Stabilizing opti. targets:** Bootstrapping est. changes after each iteration, leading to stability issues. **DQN** updates NN used for bootstrapping infrequently and maintains const. opti. target across multiple episodes. E.g. clone: hanging/online NN and fixed/target NN.
- **Max. bias:** Estimates $Q^\star$ are noisy (biased) estimates of $q^\star$. **DDQN**: instead of picking optimal action w.r.t. old network, it picks w.r.t. new network:

Policy val. fn.: measures discounted payoff of policy:
$J(\pi)\doteq\mathbb{E}_\pi[G_0]=\mathbb{E}_\pi[\sum_{t=0}^\infty\gamma^tR_t]$, and bounded ver.:
$J_T(\pi)\doteq\mathbb{E}_\pi[G_{0:T}]=\mathbb{E}_\pi[\sum_{t=0}^{T-1}\gamma^tR_t]$. Non-convex.

Score grad. est.: $\nabla_\varphi\mathbb{E}_{\tau\sim\Pi_\varphi}[G_{0:T}]=\mathbb{E}[G_{0:T}\nabla_\varphi\log\Pi_\varphi(\tau)]$
Typically var. of est. is very large. Reduce w. **baselines**:
$\mathbb{E}[G_{0:T}\nabla_\varphi\log\Pi_\varphi(\tau)]=\mathbb{E}[(G_{0:T}-b)\nabla_\varphi\log\Pi_\varphi(\tau)]$

Algorithm 12.8: REINFORCE algorithm

initialize policy weights $\boldsymbol{\varphi}$
repeat
  generate an episode (i.e., rollout) to obtain trajectory τ
  for $t=0$ **to** $T-1$ **do**
    set $g_{t:T}$ to the downstream return from time t
    $\boldsymbol{\varphi}\leftarrow\boldsymbol{\varphi}+\eta\gamma^tg_{t:T}\nabla_{\boldsymbol{\varphi}}\log\pi_{\boldsymbol{\varphi}}(\boldsymbol{a}_t\mid\boldsymbol{x}_t)$
until converged

On-policy, Model-free

- SGD with score grad est. and downstream returns.
- Not guaranteed to find an optimal policy. Can get stuck in local optima even for very small domains.

Advantage fn.: $a^\pi(\mathbf{x},\mathbf{a})\doteq q^\pi(\mathbf{x},\mathbf{a})-v^\pi(\mathbf{x})$
$=q^\pi(\mathbf{x},\mathbf{a})-\mathbb{E}_{\mathbf{a}'\sim\pi(\mathbf{x})}[q^\pi(\mathbf{x},\mathbf{a}')]$
- π is optimal $\Longleftrightarrow\forall\mathbf{x}\in\mathcal{X},\mathbf{a}\in\mathcal{A}:a^\pi(\mathbf{x},\mathbf{a})\leq0$

Policy gradient theorem: Max. $J(\varphi)$ corresponds to incr. the prob. of actions with large and decr. the prob. of actions with small value, taking into account how often the resulting policy visits certain states.
$\nabla_\varphi J(\varphi)=\sum_{t=0}^\infty\mathbb{E}_{\mathbf{x}_t,\mathbf{a}_t}[\gamma^tq^{\pi_\varphi}(\mathbf{x}_t,\mathbf{a}_t)\nabla_\varphi\pi_\varphi(\mathbf{a}_t,\mathbf{x}_t)]$

Algorithm 12.11: Online actor-critic

initialize parameters $\boldsymbol{\varphi}$ and $\boldsymbol{\theta}$
repeat
  use $\pi_{\boldsymbol{\varphi}}$ to obtain transition $(\boldsymbol{x},\boldsymbol{a},r,\boldsymbol{x}')$ and the next a
  $\boldsymbol{a}'\sim\pi_{\boldsymbol{\varphi}}(\cdot\mid\boldsymbol{x}')$
  $\delta=r+\gamma Q(\boldsymbol{x}',\boldsymbol{a}';\boldsymbol{\theta})-Q(\boldsymbol{x},\boldsymbol{a};\boldsymbol{\theta})$
  // actor update
  $\boldsymbol{\varphi}\leftarrow\boldsymbol{\varphi}+\eta Q(\boldsymbol{x},\boldsymbol{a};\boldsymbol{\theta})\nabla_{\boldsymbol{\varphi}}\log\pi_{\boldsymbol{\varphi}}(\boldsymbol{a}\mid\boldsymbol{x})$
  // critic update
  $\boldsymbol{\theta}\leftarrow\boldsymbol{\theta}+\eta\delta\nabla_{\boldsymbol{\theta}}Q(\boldsymbol{x},\boldsymbol{a};\boldsymbol{\theta})$
until converged

On-policy, Online, Model-free

- Use SARSA for learning critic; SGD for gradient est.
- Actor is not guaranteed to improve
- Typically rely on randomization in policy to encourage exploration
- Typically low sample efficiency (improve. w. TRPO)

Reduce var. of the grad estimates version with **A2C**:
$\boldsymbol{\varphi}\leftarrow\boldsymbol{\varphi}+\eta_t(Q(\boldsymbol{x},\boldsymbol{a};\boldsymbol{\theta})-V(\boldsymbol{x};\boldsymbol{\theta}))\nabla_{\boldsymbol{\varphi}}\log\pi_{\boldsymbol{\varphi}}(\boldsymbol{a}\mid\boldsymbol{x})$
$=\boldsymbol{\varphi}+\eta_tA(\boldsymbol{x},\boldsymbol{a};\boldsymbol{\theta})\nabla_{\boldsymbol{\varphi}}\log\pi_{\boldsymbol{\varphi}}(\boldsymbol{a}\mid\boldsymbol{x})$ w. Bootstrap est. $\mathbf{a}^\pi\varphi$

- **TRPO:** KL-constraint forces monotonic improvement via trust regions (on-pol., m-free, pol-gradient) • **PPO:** Clipped obj. stabilizes training by limiting update size (on-pol., m-free, AC) • **GRPO:** Group-relative advantages remove need for separate critic (on-pol., m-free, critic-less) • **DDPG:** Det. grads. enable cont. control with target networks, inject Gaussian noise "dithering" (off-pol., m-free, AC) • **SAC:** Max. entropy obj. balances exploration and performance (off-pol., m-free) • **DPO:** No explicit reward modeling (offline, m-free, ref.-based) • **PETS:** Ensembles plus MPC handle uc. in planning. (m-based) • **UCRL:** selects optimistic MDPs for efficient exploration (m-based, exploration, o.i.t.f.o.uc) • **H-UCRL:** hallucinated transitions ensure safe, robust exploration. (m-based, opt. o.i.t.f.o.uc, hierarchical)